A1

Part 1 : Shortest Distance from Point to function

a) Analytical Approach

Given: $y = x^2 + 5$ (0,0) (-4,0) (-8,0) (2,0) (6,0)

① $f(x) = x^2+5$
$f'(x) = 2x$

② $D(x) = (x-x_0)^2 + (f(x)-y_0)^2$
$D'(x) = 2(x-x_0) + 2(f(x)-y_0)f'(x) = 0$

for all points $y_0 = 0$

$0 = 2(x-x_0) + 2f(x)f'(x)$
$0 = 2[(x-x_0) + f(x)f'(x)]$
$0 = (x-x_0) + (x^2+5)(2x)$
$0 = x - x_0 + 2x^3 + 10x$
$0 = 2x^3 + 11x - x_0$

③ sub in $x_0 = -4$ (for point (-4,0))
$0 = 2x^3 + 11x + 4$
$x^* = -0.35546$ ↘ solve eqn for $x = x^*$
$f(x) = y = x^2+5$
$= 5.12635$

∴ point on curve closest to line is (-0.355, 5.126)

④ Calculate min distance
$d = \sqrt{(x^*-x_0)^2 + (y^*-y_0)^2}$
$= \sqrt{(-0.355+4)^2 + (5.126-0)^2}$
$= 6.2898$

Summary:

(x_0, y_0)	equation	(x^*, y^*)	dmin
(0,0)	$2x^3+11x = 0$	(0, 5)	5
(-4,0)	$2x^3+11x+4 = 0$	(-0.355, 5.126)	6.289
(-8,0)	$2x^3+11x+8 = 0$	(-0.672, 5.451)	9.133
(2,0)	$2x^3+11x-2 = 0$	(0.1807, 5.0326)	5.351
(6,0)	$2x^3+11x-6 = 0$	(0.5199, 5.2702)	7.603

b) Numerical Approach

1) Newton-Raphson

$f(x) = x^2+5$ For point (-4,0)
$f'(x) = 2x$

$D(x) = (x-x_0)^2 + (f(x)-y_0)^2$
$D'(x) = 2(x-x_0) + 2(f(x)-y_0)f'(x)$
$= 2(x+4) + 2(x^2+5-0)(2x)$
$= 2x+8 + (2x^2+10)(2x)$
$= 2x+8 + 4x^3+20x$
$= 4x^3 + 22x + 8$
$D''(x) = 12x^2 + 22$

$x_{n+1} = x_n - \frac{D'(x_n)}{D''(x_n)}$
$= x_n - \frac{4x^3+22x+8}{12x^2+22}$

Initial guess $x_0 = 0$

Iteration #1:
$x_0 = 0$
$D'(x) = 8$ $D''(x) = 22$
$x_{n+1} = 0 - \frac{8}{22}$
$x_1 = -\frac{8}{22} = -0.3636$

Iteration #2:
$x_1 = -0.3636$
$D'(x) = -0.1923$ $D''(x) = 23.5867$
$x_2 = -0.3636 - \frac{-0.1923}{23.5867}$
$x_2 = -0.3494$

Iteration #3:
$x_2 = -0.3494$
$D'(x) = 0.14155$ $D''(x) = 23.465$
$x_3 = -0.3494 - \frac{0.14155}{23.465}$
$= -0.35543$

Iteration #4:
$x_3 = -0.35543$
$D'(x) = 0.000877$ $D''(x) = 23.5157$
$x_4 = -0.35543 - \frac{0.000877}{23.5157}$
$= -0.35433$

∴ $x^* = -0.35433$ $y^* = 5.1255$
$D(x^*) = \sqrt{(x^*-x_0)^2 + (y^*-y_0)^2}$
$= \sqrt{(-0.35433+4)^2 + 5.1255^2}$
$= 6.28984 \approx d_{min}$

Point (0,0):

$D'(x) = 2(x-x_0) + 2(f(x)-y_0)f'(x)$ $\quad D''(x) = 12x^2+22$

$= 2x + 2(x^2+5)(2x)$

$= 2x + 4x^3 + 20x$

$= 4x^3 + 22x$

Iteration #1:

initial guess $x_0 = 0$

$D'(0) = 0 \quad D''(0) = 22$

$x_1 = 0 - \frac{0}{22}$

$= 0$

Iteration #2:

$D'(0) = 0 \quad D''(0) = 22$

$x_2 = 0 - \frac{0}{22}$

$= 0$

$(0,5)$

Point: (2,0)

$D'(x) = 2(x-x_0) + 2(f(x)-y_0)f'(x)$ $\quad D''(x) = 12x^2+22$

$= 2(x-2) + 2(x^2+5)(2x)$

$= 2x - 4 + 4x^3 + 20x$

$= 4x^3 + 22x - 4$

Iteration #1

$x_0 = 0$

$D'(0) = -4 \quad D''(0) = 22$

$x_1 = 0 - \left(\frac{-4}{22}\right)$

$= 2/11$

Iteration #2

$D'(2/11) = \frac{32}{1331} \quad D''(2/11) = \frac{2710}{121}$

$x_2 = \frac{2}{11} - \frac{32/1331}{2710/121}$

$= 0.1807447..$

$(0.18, 5.03)$

2) Golden Section Search

$D(x) = (x-x_0)^2 + (y-y_0)^2$

for point (-4,0)

$D(x) = (x+4)^2 + y^2$

$= (x+4)^2 + (x^2+5)^2$

Interval: $[-1,0] \Leftarrow [a,b]$

$D(-1) = (3)^2 + (1+5)^2 \quad D(-0.5) = (3.5)^2 + (5.25)^2 \quad D(0) = 4^2 + 5^2$

$= 45 \qquad = 39.8125 \qquad = 41$

↑ this is less than end points so includes the minimum

$\phi = \frac{1+\sqrt{5}}{2} \quad r = 2 - \phi$

$r = 2 - \left(\frac{1+\sqrt{5}}{2}\right)$

$r = \frac{3-\sqrt{5}}{2}$

Iteration #1:

$x_1 = a + r(b-a) \quad x_2 = b - r(b-a)$

$= -1 + \left(\frac{3-\sqrt{5}}{2}\right)(0+1) \quad = 0 - \left(\frac{3-\sqrt{5}}{2}\right)(0+1)$

$= \frac{1-\sqrt{5}}{2} \approx -0.618034 \quad = \frac{\sqrt{5}-3}{2} \approx -0.381966$

$D(x_1) = 40.40325 \quad D(x_2) = 39.570436$

$D(x_2) < D(x_1)$

New interval: $[x_1, b]$

Iteration #2: $[-0.618034, 0]$

$x_1 = -0.618034 + \left(\frac{3-\sqrt{5}}{2}\right)(0.618034) \quad x_2 = 0 - \left(\frac{3-\sqrt{5}}{2}\right)(0.618034)$

$= -0.381966 \quad = -0.236068$

$D(x_1) = 39.12243 \quad D(x_2) = 39.72757$

$D(x_2) > D(x_1)$

New interval: $[a, x_2]$

Iteration #3: $[-0.618034, -0.236068]$

$x_1 = -0.472135 \quad x_2 = -0.381966$

$D(x_1) = 39.72464 \quad D(x_2) = 39.5704$

$D(x_2) < D(x_1)$

New interval: $[x_1, b]$

Iteration #4: $[-0.472135, -0.236068]$

$x_1 = -0.381965 \quad x_2 = -0.32623$

$D(x_1) = 39.5704 \quad D(x_2) = 39.512167$

$D(x_2) > D(x_1)$

New interval: $[a, x_2]$

Iteration #5: $[-0.472135, -0.32623]$

$x_1 = -0.416404 \quad x_2 = -0.3819607$

$D(x_1) = 39.60614 \quad D(x_2) = 39.57043$

x^* = midpoint btwn x_1 & x_2

$x^* = \frac{-0.416404 + (-0.3819607)}{2} = -0.39918$ $\quad (-0.399, 5.16)$

$D(x^*) = 39.58474$

$d_{min} = \sqrt{D(x^*)} = 6.29164\ldots$

Point (-8,0):

$D'(x) = 2(x-x_0) + 2(f(x)-y_0)f'(x)$ $\quad D''(x) = 12x^2+22$

$= 2(x+8) + 2(x^2+5)(2x)$

$= 2x + 16 + 4x^3 + 20x$

$= 4x^3 + 22x + 16$

Iteration #1

$x_0 = 0$

$D'(0) = 16 \quad D''(0) = 22$

$x_1 = 0 - \frac{16}{22}$

$= 8/11$

Iteration #2

$D'(8/11) = 33.5386 \quad D''(8/11) = 28.3471$

$x_2 = \frac{8}{11} - \frac{33.5386}{28.3471}$

$= -0.45586$

$(-0.46, 5.21)$

Point (6,0):

$D'(x) = 2(x-x_0) + 2(f(x)-y_0)f'(x)$ $\quad D''(x) = 12x^2+22$

$= 2(x-6) + 2(x^2+5)(2x)$

$= 2x - 12 + 4x^3 + 20x$

$= 4x^3 + 22x - 12$

Iteration #1

$x_0 = 0$

$D'(0) = -12 \quad D''(0) = 22$

$x_1 = 0 - \left(\frac{-12}{22}\right)$

$= 6/11$

Iteration #2

$D'(6/11) = 0.649135 \quad D''(6/11) = \frac{3094}{121}$

$x_2 = \frac{6}{11} - \frac{0.649135}{3094/121}$

$= 0.52006$

$(0.52, 5.27)$

Point (0,0):

$D(x) = x^2 + (x^2+5)^2$

Interval: $[-1, 0.5]$

Iteration #1:

$x_1 = a + r(b-a)$
$= -1 + \left(\frac{3-\sqrt{5}}{2}\right)(0.5-(-1))$
$= \frac{5-3\sqrt{5}}{4} \approx -0.42705$

$x_2 = b - r(b-a)$
$= 0.5 - \left(\frac{3-\sqrt{5}}{2}\right)(0.5+1)$
$= \frac{-7+3\sqrt{5}}{4} \approx -0.07295$

$D(x_1) = 27.0393$

$D(x_2) = 25.0585$

$D(x_1) > D(x_2)$

New interval $[x_1, b]$

Iteration #2:

$[-0.42705,\ 0.5]$

$x_1 = \frac{-13+5\sqrt{5}}{4} \approx -0.4549$ $\quad x_2 = \frac{7-3\sqrt{5}}{2} \approx 0.146898$

$D(x_1) = 27.3192$ $\quad D(x_2) = 25.2346$

$D(x_1) > D(x_2)$

new interval $[x_1, b]$

$\Rightarrow [-0.45, 0.5]$

Point (-8,0):

$D(x) = (x+8)^2 + (x^2+5)^2$

Iteration #1: $[-1,0]$

$x_1 = \frac{1-\sqrt{5}}{2} \approx -0.618034$ $\quad x_2 = \frac{\sqrt{5}-3}{2} \approx -0.381966$

$D(x_1) = 83.4589$ $\quad D(x_2) = 84.5147$

$D(x_1) < D(x_2)$

$\Rightarrow [a, x_2]$

Iteration #2: $[-1, -0.38196]$

$x_1 = -3+\sqrt{5} \approx -0.76393$ $\quad x_2 = \frac{1-\sqrt{5}}{2} \approx -0.61803$

$D(x_1) = 83.537$ $\quad D(x_2) = 83.4589$

$D(x_1) > D(x_2)$

$\Rightarrow [x_1, b]$

$\Rightarrow [-0.76, -0.38]$

Point (2,0)

$D(x) = (x-2)^2 + (x^2+5)^2$

Iteration #1: $[0,1]$

$x_1 = \frac{3-\sqrt{5}}{2} \approx 0.381966$ $\quad x_2 = \frac{-1+\sqrt{5}}{2} \approx 0.61803$

$D(x_1) = 29.0983$ $\quad D(x_2) = 51-9\sqrt{5} \approx 30.876$

$D(x_1) < D(x_2)$

$\Rightarrow [a, x_2]$

Iteration #2: $[0, 0.61803]$

$x_1 = -2+\sqrt{5} \approx 0.23607$ $\quad x_2 = \frac{3-\sqrt{5}}{2} \approx 0.38196$

$D(x_1) = 28.6718$ $\quad D(x_2) = 29.098$

$D(x_1) < D(x_2)$

$\Rightarrow [a, x_2]$

$\Rightarrow [0, 0.382]$

Point (6,0)

$D(x) = (x-6)^2 + (x^2+5)^2$

Iteration #1: $[0,1]$

$x_1 = \frac{3-\sqrt{5}}{2} \approx 0.381966$ $\quad x_2 = \frac{-1+\sqrt{5}}{2} \approx 0.61803$

$D(x_1) = 58.0426$ $\quad D(x_2) = 87-13\sqrt{5} \approx 57.9311$

$D(x_1) > D(x_2)$

$\Rightarrow [x_1, b]$

Iteration #2: $[0.38196, 1]$

$x_1 = \frac{-1+\sqrt{5}}{2} \approx 0.61803$ $\quad x_2 = 3-\sqrt{5} \approx 0.7639$

$D(x_1) = 87-13\sqrt{5} \approx 57.9311$ $\quad D(x_2) = 58.5929$

$D(x_1) < D(x_2)$

$\Rightarrow [a, x_2]$

$\Rightarrow [0.382, 0.76]$

Part 2 : Fitting to an Equation

a) Analytical Approach

Given points $(0,0.5)$ $(2,3.5)$ $(1,1.5)$ $(3,7.5)$

x_i	y_i
0	0.5
1	1.5
2	3.5
3	7.5

$n=4$ $\Rightarrow$ $\Sigma x_i = 6$ $\quad \Sigma x_i^2 = 14$ $\quad \Sigma x_i y_i = 31$

$\Sigma y_i = 13$ $\quad \Sigma y_i^2 = 71$

1) LINE: $\hat{y} = mx + b$

$e_i = \hat{y}_i - y_i$

divide by 4 bc 4 data points

$$MSE(m,b) = \frac{1}{4}\sum_{i=1}^{4}(mx_i + b - y_i)^2$$

$$= \frac{1}{4}\left[(m(0)+b-0.5)^2 + (m(1)+b-1.5)^2 + (m(2)+b-3.5)^2 + (m(3)+b-7.5)^2\right]$$

$$= \frac{1}{4}\left[(b-0.5)^2 + (m+b-1.5)^2 + (2m+b-3.5)^2 + (3m+b-7.5)^2\right]$$

$$= \frac{1}{4}\sum_{i=1}^{4}(mx_i+b-y_i)(mx_i+b-y_i)$$

$$= \frac{1}{4}\sum_{i=1}^{4} m^2x_i^2 + mbx_i - mx_iy_i + mbx_i + b^2 - by_i - mx_iy_i - by_i + y_i^2$$

$$= \frac{1}{4}\left[14m^2 + 6mb - 31m + 6mb + 4b^2 - 13b - 31m - 13b + 71\right)$$

$$= \frac{1}{4}\left(14m^2 + 12mb - 62m + 4b^2 - 26b + 71\right)$$

$$= 3.5m^2 + 3mb - 15.5m + b^2 - 6.5b + 17.75$$

$$\frac{\partial MSE}{\partial m} = 7m + 3b - 15.5 = 0$$

$$7m + 3b = 15.5$$

$$14m + 6b = 31$$

$$-\ 9m + 6b = 19.5$$

$$5m = 11.5$$

$$m = 2.3$$

$$\frac{\partial MSE}{\partial b} = 3m + 2b - 6.5 = 0$$

$$3m + 2b = 6.5$$

$$6m + 4b = 13$$

$$9m + 6b = 19.8$$

$$b = \frac{19.5 - 9m}{6}$$

$$= \frac{19.5 - 9(2.3)}{6}$$

$$= -0.2$$

$\therefore y = 2.3x - 0.2$

MSE:

$$SSE = (-0.2-0.5)^2 + (2.1-1.5)^2 + (4.4-3.5)^2 + (6.7-7.5)^2$$

$$= 2.3$$

$$MSE = \frac{SSE}{4} = 0.575$$

2) PARABOLA : $\hat{y} = ax^2 + bx + c$

x_i	y_i
0	0.5
1	1.5
2	3.5
3	7.5

$\sum_{i=1}^{4} x_i^2 = 14$ $\quad \sum_{i=1}^{4} x_i^3 = 36$ $\quad \sum_{i=1}^{4} x_iy_i = 31$ $\quad \sum x_i = 6$

$\sum_{i=1}^{4} y_i^2 = 71$ $\quad \sum_{i=1}^{4} x_i^4 = 98$ $\quad \sum_{i=1}^{4} x_i^2y_i = 83$ $\quad \sum y_i = 13$

$$\theta = \hat{y}_i - y_i$$

$$MSE(a,b,c) = \frac{1}{4}\sum_{i=1}^{4}(ax_i^2 + bx_i + c - y_i)^2$$

$$= \frac{1}{4}\sum_{i=1}^{4}(ax_i^2+bx_i+c-y_i)(ax_i^2+bx_i+c-y_i)$$

$$= \frac{1}{4}\sum_{i=1}^{4} a^2x_i^4 + abx_i^3 + acx_i^2 - ax_i^2y_i + abx_i^3 + b^2x_i^2 + bcx_i - bx_iy_i + acx_i^2 + bcx_i + c^2 - cy_i$$

$$- ax_i^2y_i - bx_iy_i - cy_i + y_i^2$$

$$= \frac{1}{4}\sum_{i=1}^{4} a^2x_i^4 + 2abx_i^3 + 2acx_i^2 - 2ax_i^2y_i + b^2x_i^2 + 2bcx_i - 2bx_iy_i + c^2 - 2cy_i + y_i^2$$

$$= \frac{1}{4}\left(98a^2 + 72ab + 28ac - 166a + 14b^2 + 12bc - 62b + 4c^2 - 26c + 71\right)$$

$$= 24.5a^2 + 3.5b^2 + c^2 + 18ab + 7ac + 3bc - 41.5a - 15.5b - 6.5c + 17.75$$

$$\frac{\partial MSE}{\partial a} = 49a + 18b + 7c - 41.5 = 0$$

$$98a + 36b + 14c = 83$$

$$98a + 36(2.3-3a) + 14\left(\frac{13-14a-6b}{4}\right) = 83$$

$$98a + 82.8 - 108a + 45.5 - 49a - 21b = 83$$

$$-59a - 21b = -45.3$$

$$45.3 = 59a + 21(2.3-3a)$$

$$45.3 = 59a + 48.3 - 63a$$

$$4a = 3$$

$$a = 0.75$$

$$\frac{\partial MSE}{\partial b} = 7b + 18a + 3c - 15.5 = 0$$

$$36a + 14b + 6c = 31$$

$$36a + 14b + 6\left(\frac{13-14a-6b}{4}\right) = 31$$

$$36a + 14b + 19.5 - 21a - 9b = 31$$

$$15a + 5b = 11.5$$

$$b = \frac{11.5 - 15a}{5}$$

$$b = 2.3 - 3a$$

$$b = 2.3 - 3(0.75)$$

$$= 0.05$$

$$\frac{\partial MSE}{\partial c} = 2c + 7a + 3b - 6.5 = 0$$

$$14a + 6b + 4c = 13$$

$$c = \frac{13 - 14a - 6b}{4}$$

$$c = \frac{13 - 14(0.75) - 6(0.05)}{4}$$

$$= 0.55$$

$\therefore \hat{y} = 0.75x^2 + 0.05x + 0.55$

MSE :

$$SSE = \left((0.55-0.5)^2 + (1.35-1.5)^2 + (3.65-3.5)^2 + (7.45-7.5)^2\right)$$

$$= (0.05)$$

$$MSE = \frac{SSE}{4} = 0.0125$$

b) Numerical Approach – Newton Raphson Method

1) LINE : $\hat{y} = mx + b$

$MSE(m,b) = 3.5m^2 + 3mb - 15.5m + b^2 - 6.5b + 17.5$

$\frac{\partial MSE}{\partial m} = 7m + 3b - 15.5$ $\quad$ $\frac{\partial MSE}{\partial b} = 3m + 2b - 6.5$

$\frac{\partial^2 MSE}{\partial m^2} = 7$ $\quad$ $\frac{\partial^2 MSE}{\partial b^2} = 2$

① set :

$m_{n+1} = m_n - \frac{7m_n + 3b - 15.5}{7}$

$= 0 - \frac{7(0) + 3b - 15.5}{7}$

$= \frac{15.5 - 3b}{7}$

$b_{n+1} = b_n - \frac{3m_{n+1} + 2b_n - 6.5}{2}$

$= 0 - \left(\frac{3m_{n+1} - 6.5}{2}\right)$

$= \frac{6.5 - 3m_{n+1}}{2}$

② choose initial values : $m_0 = 0$ $\quad b_0 = 0$

Iteration #1

$m_1 = \frac{15.5 - 3(0)}{7}$ $\quad$ $b_1 = \frac{6.5 - 3m_1}{2}$

$= 2.214285..$ $\quad$ $= \frac{6.5 - 3(2.214285..)}{2}$

$= -0.0714285..$

$\hat{y} = 2.214x - 0.0714$

$SSE = (-0.0714 - 0.05)^2 + (2.1426 - 1.5)^2 + (4.3566 - 3.5)^2 + (6.5706 - 7.5)^2$

$= 2.02522$

$MSE = \frac{SSE}{4} = 0.50630..$

③ Iteration #2

$m_2 = \frac{15.5 - 3b_1}{7}$ $\quad$ $b_2 = \frac{6.5 - 3(2.2448898)}{2}$

$= \frac{15.5 - 3(-0.071429)}{7}$ $\quad$ $= -0.11734$

$= 2.2448898$

$\hat{y} = 2.24x - 0.117$

$SSE = (-0.117 - 0.5)^2 + (2.123 - 1.5)^2 + (4.363 - 3.5)^2 + (6.603 - 7.5)^2$

$= 2.318196$

$MSE = \frac{SSE}{4} = 0.5795549..$

2) PARABOLA : $\hat{y} = ax^2 + bx + c$

$MSE(a,b,c) = 24.5a^2 + 3.5b^2 + c^2 + 18ab + 7ac + 3bc - 41.5a - 15.5b - 6.5c + 17.75$

$\frac{\partial MSE}{\partial a} = 49a + 18b + 7c - 41.5$ $\quad$ $\frac{\partial MSE}{\partial b} = 7b + 18a + 3c - 15.5$ $\quad$ $\frac{\partial MSE}{\partial c} = 2c + 7a + 3b - 6.5$

$\frac{\partial^2 MSE}{\partial a^2} = 49$ $\quad$ $\frac{\partial^2 MSE}{\partial b^2} = 7$ $\quad$ $\frac{\partial^2 MSE}{\partial c^2} = 2$

$a_{n+1} = a_n - \frac{49a + 18b + 7c - 41.5}{49}$

$= \frac{41.5 - 18b_n - 7c_n}{49}$

$b_{n+1} = b_n - \frac{7b + 18a + 3c - 15.5}{7}$

$= \frac{15.5 - 18a_{n+1} - 3c_n}{7}$

$c_{n+1} = c_n - \frac{2c + 7a + 3b - 6.5}{2}$

$= \frac{6.5 - 7a_{n+1} - 3b_{n+1}}{2}$

① initial params :

$a_0 = 0$ $\quad b_0 = 0$ $\quad c_0 = 0$ $\quad \Rightarrow \hat{y} = 0$

Iteration #1 :

$a_1 = \frac{41.5 - 18(0) - 7(0)}{49}$ $\quad$ $b_1 = \frac{15.5 - 18(0.8469) - 3(0)}{7}$

$= 0.8469387..$ $\quad$ $= 0.036443..$

$c_1 = \frac{6.5 - 7(0.8469) - 3(0.036443)}{2}$

$= 0.231185..$

$\therefore \hat{y} = 0.847x^2 + 0.036x + 0.231$

MSE :

$SSE = (0.231 - 0.5)^2 + (1.114 - 1.5)^2 + (3.691 - 3.5)^2 + (7.962 - 7.5)^2$

$= 0.450281$

$MSE = \frac{SSE}{4} = 0.11257$

Iteration #2 :

$a_1 = \frac{41.5 - 18(0.036) - 7(0.231)}{49}$ $\quad$ $b_1 = \frac{15.5 - 18(0.8007) - 3(0.231)}{7}$

$= 0.800714$ $\quad$ $= 0.056342$

$c_1 = \frac{6.5 - 7(0.800714) - 3(0.056342)}{2}$

$= 0.363012..$

$\hat{y} = 0.801x^2 + 0.056x + 0.363$

MSE :

$SSE = (0.363 - 0.5)^2 + (1.22 - 1.5)^2 + (3.679 - 3.5)^2 + (7.74 - 7.5)^2$

$= 0.19681$

$MSE = \frac{SSE}{4} = 0.0492$